

torch.jit.annotate#

<https://pytorch.org/docs/stable/generated/torch.jit.annotate.html#torch.jit.annotate>

Description:

Use to give type of the_value in TorchScript compiler. This method is a pass-through function that returns the_value, used to hint TorchScript compiler the type of the_value. It is a no-op when running outside of TorchScript. Though TorchScript can infer correct type for most Python expressions, there are some cases where type inference can be wrong, including: Empty containers like [] and {}, which TorchScript assumes to be container of Tensor Optional types like Optional[T] but assigned a valid value of type T, TorchScript would assume it is type T rather than Optional[T]

Function Signature

```
torch.jit.annotate(the_type, the_value)[source]#
```

Parameters

Returns

Returns:

the_value is passed back as return value.

Code Examples

```
import torch
from typing import Dict

@torch.jit.script
def fn():
    # Telling TorchScript that this empty dictionary is a (str -
    > int) dictionary
    # instead of default dictionary type of (str -> Tensor).
    d = torch.jit.annotate(Dict[str, int], {})

    # Without `torch.jit.annotate` above, following statement
    would fail because of
    # type mismatch.
    d["name"] = 20
```

Detailed Documentation

Documentation

Use to give type of the_value in TorchScript compiler.

This method is a pass-through function that returns the_value, used to hint TorchScript compiler the type of the_value. It is a no-op when running outside of TorchScript.

Though TorchScript can infer correct type for most Python expressions, there are some cases where type inference can be wrong, including:

Empty containers like [] and {}, which TorchScript assumes to be container of Tensor

Optional types like Optional[T] but assigned a valid value of type T, TorchScript would assume it is type T rather than Optional[T]

Note that `annotate()` does not help in `__init__` method of `torch.nn.Module` subclasses because it is executed in eager mode. To annotate types of `torch.nn.Module` attributes, use `Attribute()` instead.

`the_type` – Python type that should be passed to TorchScript compiler as type hint for `the_value`

`the_value` – Value or expression to hint type for.

`the_value` is passed back as return value.